

From Psychology to Neurobiology

For reasons we just saw, the laws of probability are said to be counterintuitive by the researchers in the cognitive and behavioral sciences. We are *probability blind*, these scientists say. This chapter will rapidly illustrate some manifestations of such blindness, with a cursory exposition of the research in that area.

The idea of probability blindness gave rise to an entire discipline dedicated to the study of the effect these biases carry on our behavior. It is filling out library shelves and causing the creation of numerous investment funds dedicated to the sister idea that people do not behave in a rational way in the markets. Some funds have been built around the idea that people overreact to news, while others have been devoted to the notion that, to the contrary, people underreact (I was told early in my career that the more diversity the better for the market). These beliefs give rise to two categories of trading strategies. On one side we find the contrarians who subscribe to the following rationale: *Hey, since people systematically overreact, let us take the other side, sell the winners and buy the losers*. On the other side stand the momentum players who do the exact opposite: *Since markets do not adjust fast enough, let us buy the winners and sell the losers*. Because of randomness, both categories will show periodic victories, which cannot prove directly that either theory is right or wrong.

Even psychiatrists and clinical psychologists are joining the fray by becoming “experts” – after all they know more about the human mind than those financial economists with their unrealistic, unscientific equations, and it is, after all, human behavior that ultimately influences the markets. A yearly conference in Boston gathers medical doctors and psychology researchers musing over market strategies. The idea may seem simple, perhaps even boring until we encounter professionals, whom we expect would deal with the matter with maximal expertise, falling right into the trap like the man on the street.

OUR NATURAL HABITAT

I will not delve too deeply into amateur evolutionary theory to probe at the reasons (besides, in spite of having spent some time in libraries I feel that I am truly an amateur in the subject matter). Clearly, the environment for which we have built our genetic endowment is not the one that prevails today. I have not told too many of my colleagues that their decision-making contains some lingering habits of cavemen – but when markets experience an abrupt move, I experience the same rush of adrenaline as if a leopard was seen prowling near my trading desk. Some of my colleagues who break telephone handles upon losing money might be even closer in their psychological makeup to our common origin.

This might be a platitude to those who frequent the Greek and Latin classics, but we never fail to be surprised when noticing that people a couple of dozen centuries removed from us can exhibit similar sensibility and feelings. What used to strike me as a child upon visiting museums is that ancient Greek statues exhibit men with traits indistinguishable from ours (only more harmonious and aristocratic). I was so wrong to believe that 2,200 years was a long time. Proust wrote frequently about the surprise people have while coming across emotions in Homeric heroes that are similar to those we experience today. By genetic standards, these Homeric heroes of 30 centuries ago in all likelihood have the exact identical genetic makeup as the pudgy middle-aged man you see schlepping groceries in the parking lot. More than that. In fact we are truly identical to the man who perhaps 80 centuries ago started being called “civilized”, in that strip of land stretching from Southeastern Syria to Southwestern Mesopotamia.

What is our natural habitat? By natural habitat, I mean the environment in which we reproduced the most, the one in which we spent the highest number of generations. The consensus among anthropologists is that we have been around as a separate species for 130,000 years, most of which were spent in the African savannah. But we do not have to go back that far in history to get the point. Imagine life in an early urban settlement, in Middle-Town, Fertile Crescent, only about 3,000 years ago – surely modern times from a genetic standpoint. Information is limited by the physical means of its transmission; one cannot travel fast, hence information will come from faraway places in